



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

PHY 1202 Lab



16/10/2025



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

LAB CLASS TEACHER

Dr. Syed Jamal Ahmed
Professor

Mr. Abdulla Al Noman
Assistant Professor



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

INSTRUCTOR

Abu Hena Md Golam Mortuza

Junior Instructor

Email: akt.duet67@gmail.co

Mr. Md. Borhan Uddin

Junior Instructor

Email: brhn@duet.ac.bd

Lab Assistant

Mr. Md. Tofazzol Hossain (Senior Lab Attendant)

Mr. Md. Monirozzaman (Lab Attendant)



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

LAB TIME

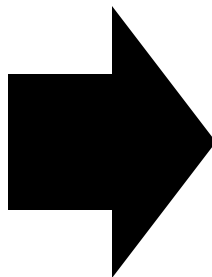
Wednesday

02:00 pm – 04:30 pm

LOCATION

Physics Laboratory (201)

Come in time





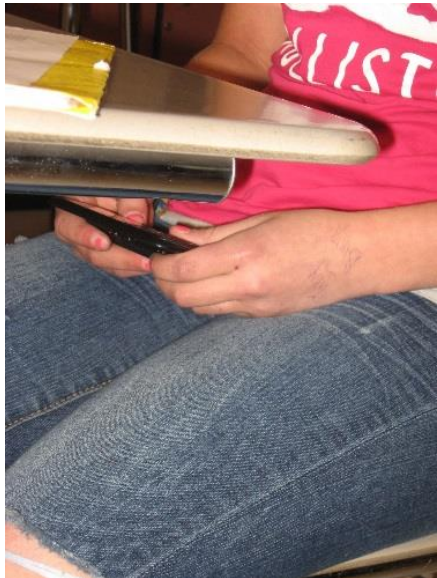
Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

CELL PHONES



**Please turn it off
before arriving
to class!**





Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

ATTENDANCE

- **10 Marks**
- **Be on Time!**
(Before 05 minutes **Mandatory**)
- Don't expect labs to end early
- Let class teacher know if you anticipate an absent
- Two time attendance
(First time- at starting and second time just before the end of the class)





Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

MANUAL

- Read lab Manual before come to Lab
(Supplied manual)
- Always follow lab procedure





Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

RULES

- ☐ Give your report to the CR within 5 minutes after starting the class **(both complete and incomplete sheet)**
- ☐ CR have to give the report to the respective teacher for signing the incomplete sheet.
- ☐ Collect the apparatus **(Mr. Tofazzal Hosen and Mr. Monirozzaman)**
- ☐ Start your experiment with a rough sheet
- ☐ Incomplete sheet will be given 60 minutes later with the sign of the respective.



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

REPORT PREPARATION

Incomplete sheet

Complete sheet

Incomplete sheet

1. Cover Page
2. Figure (In the Left side of the page)
3. Theory
4. Apparatus
5. Experimental data/table (Keep blank)



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

EXAMPLE OF INCOMPLETE SHEET

1. Cover Page

	Department of Physics						
	Dhaka University of Engineering and Technology, Gazipur-1707						
Course Title: General Physics Sessional							
Course Code: Phy 1102							
Date of Allocation:	D	M	Y	Date of Submission:	D	M	Y
Experiment No.:							
Name of the Experiment:							
<u>Submitted by</u>				<u>Submitted to</u>			
Name:				1. Dr. Md. Sahab Uddin			
Department:				Professor,			
Student ID:				Dept. of Physics DUET			
Group No:				2. Dr. Farah Deeba			
Year / Semester:				Associate Professor,			
Session:				Dept. of Physics, DUET			
Marks Allocated				Marks Obtained			
12.5							



2. Figure:

Figure:

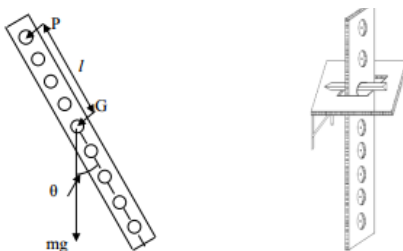


Fig. 1: Compound pendulum

3. Theory:

Name of the experiment:

TO DETERMINE THE VALUE OF 'g' ACCELERATION DUE TO GRAVITY BY MEANS OF A COMPOUND PENDULUM

Theory:

A rigid body of any shape which can swing in a vertical plane about a horizontal axis passing through it is called a compound or physical pendulum.

In Fig. 1 a compound pendulum is pivoted about a horizontal frictionless axis through P and is displaced from its equilibrium position by an angle θ . In the equilibrium position the center of gravity G of the body is vertically below P. The distance GP is l and the mass of the body is m . The restoring torque for an angular displacement θ is

$$\tau = -mgl \sin \theta \dots\dots\dots (1)$$

for small amplitudes, $\sin \theta \approx \theta$

$$I \frac{d^2 \theta}{dt^2} = -mgl \theta \dots\dots\dots (2)$$

where, I is the moment of inertia of the body through the axis P. Expression (2) represents a simple harmonic motion and hence the time period of oscillation is given by

$$T = 2\pi \sqrt{\frac{I}{mgl}} \dots\dots\dots (3)$$

If k is the radius of gyration of the pendulum about an axis through G parallel to the axis of oscillation through P, from the parallel axes theorem, $I = m(k^2 + l^2)$ and so

$$T = 2\pi \sqrt{\frac{k^2 + l^2}{gl}} \dots\dots\dots (4)$$

Equation 1 can be written in the quadratic form

$$l^2 - \frac{T^2 g}{4\pi^2} l + k^2 = 0 \dots\dots\dots (5)$$

If l_1 and l_2 are the two roots of equation (5) for a given time period T , then we can write

$$l_1 + l_2 = \frac{T^2 g}{4\pi^2} \dots\dots\dots (6)$$

$$l_1 \cdot l_2 = k^2 \dots\dots\dots (7)$$

arranged to get

$$g = \frac{4\pi^2 (l_1 + l_2)}{T^2} \dots\dots\dots (8)$$

Obviously, $l_1 + l_2$ is the length of the equivalent simple pendulum. Thus g can be calculated using Eq. 8.

Apparatus:

Compound pendulum, Stop-watch, Metre scale and, Balance with weight box

4. Apparatus



5. Experimental Data/Table:

Experimental Data:

Table 1: Data for time period, T

End	Hole number	Distance from C.G., l (cm)	Time for 20 oscillations, t (sec)	Time period T (sec)
A	1			
	2			
	3			
	4			
	5			
	6			
	7			
	8			
	9			
	10			
B	1			
	2			
	3			
	4			
	5			
	6			
	7			
	8			
	9			
	10			

Table 2: Data from $T-l$ curve

Time period, T_1 (sec)	$l_1 + l_2 = (ac + bd)/2$ (cm)	g_1 (cm/sec)	Time period, T_2 (sec)	$l_1 + l_2 = (d'e' + b'd)/2$ (cm)	g_2 (cm/sec)	Mean g (cm/sec)



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

Complete sheet

1. Cover Page
2. Figure
3. Theory
4. Apparatus
5. Experimental data/table
6. Calculations
7. Percentage of error
8. Results
9. Discussion



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

EXAMPLE OF COMPLETE SHEET

1. Cover Page



Department of Physics

Dhaka University of Engineering and Technology,
Gazipur-1707

Course Title: Physics Sessional-II

Course Code: Ph-2101

Date of Allocation:	D	M	Y	Date of Submission:	D	M	Y
	03	05	2017				

Experiment No.: 01

Name of the Experiment: **Determination of the value of acceleration due to gravity (g) by means of a compound pendulum**

Submitted by

Submitted to

Name: **Mutasim Musad**

Department: Civil Engineering

Student ID No: **155001**

Group No: **02**

Year / Semester: 2nd / 1st

Session: 2016-2017

1. Dr. A. T. M. Kaosar Jamil
Professor

2. Dr. Md. Rasadujjaman
Assistant Professor

Marks Allocated	Marks Obtained
12.5	



2. Figure:

Figure:

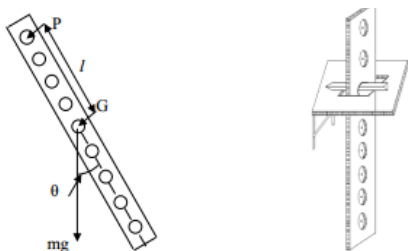


Fig. 1: Compound pendulum

3. Theory:

Name of the experiment:

TO DETERMINE THE VALUE OF 'g' ACCELERATION DUE TO GRAVITY BY MEANS OF A COMPOUND PENDULUM

Theory:

A rigid body of any shape which can swing in a vertical plane about a horizontal axis passing through it is called a compound or physical pendulum.

In Fig.1 a compound pendulum is pivoted about a horizontal frictionless axis through P and is displaced from its equilibrium position by an angle θ . In the equilibrium position the center of gravity G of the body is vertically below P. The distance GP is l and the mass of the body is m . The restoring torque for an angular displacement θ is

$$\tau = -mgl \sin \theta \dots \dots \dots (1)$$

for small amplitudes, $\sin \theta \approx \theta$

$$I \frac{d^2 \theta}{dt^2} = -mgl \theta \dots \dots \dots (2)$$

where, I is the moment of inertia of the body through the axis P. Expression (2) represents a simple harmonic motion and hence the time period of oscillation is given by

$$T = 2\pi \sqrt{\frac{I}{mgl}} \dots \dots \dots (3)$$

If k is the radius of gyration of the pendulum about an axis through G parallel to the axis of oscillation through P, from the parallel axes theorem, $I = m(k^2 + l^2)$ and so

$$T = 2\pi \sqrt{\frac{k^2 + l^2}{gl}} \dots \dots \dots (4)$$

Equation 1 can be written in the quadratic form

$$l^2 - \frac{T^2 g}{4\pi^2} l + k^2 = 0 \dots \dots \dots (5)$$

If l_1 and l_2 are the two roots of equation (5) for a given time period T , then we can write

$$l_1 + l_2 = \frac{T^2 g}{4\pi^2} \dots \dots \dots (6)$$

$$l_1 \cdot l_2 = k^2 \dots \dots \dots (7)$$

Equation 6 can be rearranged to get

$$g = \frac{4\pi^2 (l_1 + l_2)}{T^2} \dots \dots \dots (8)$$

Obviously, $l_1 + l_2$ is the length of the equivalent simple pendulum. Thus g can be calculated using Eq. 8.

Apparatus:

Compound pendulum, Stop-watch, Metre scale and, Balance with weight box

4. Apparatus



6. Calculations:

Calculations:

Data from graph,

$$(i) l_1 + l_2 = (ac + bd)/2 = 24 \quad \text{cm,}$$

$$(ii) t_1 + t_2 = (d't' + b'd)/2 = 22 \quad \text{cm}$$

$$g_1 = \frac{4\pi^2(l_1 + l_2)}{T_1^2} = 980 \quad \text{cm/sec,}$$

$$g_2 = \frac{4\pi^2(t_1' + t_2')}{T_2^2} = 970 \quad \text{cm/sec}$$

$$\text{Hence, } g = (g_1 + g_2)/2 = 975 \quad \text{cm/sec}$$

Expected value or expected value = 980 cm/sec

$$\text{Percentage of error} = \frac{\text{Expected value} - \text{Experimental value}}{\text{Expected value}} \times 100 \% = 4 \%$$

$$\text{Percentage of error} = \frac{\text{Expected value} - \text{Experimental value}}{\text{Expected value}} \times 100 \% = 4 \%$$

5. Experimental Data/Table:

Experimental Data:

Table 1: Data for time period, T

End	Hole number	Distance from C.G., l (cm)	Time for 20 oscillations, t (sec)	Time period T (sec)
A	1	50	38	1.55
	2	45	45	1.53
	3	40	37	1.25
	4	35	42	1.55
	5	30	38	1.53
	6	25	45	1.25
	7	20	37	1.55
	8	15	42	1.53
	9	10	38	1.25
	10	05	45	1.55
B	1	50	37	1.53
	2	45	42	1.25
	3	40	38	1.55
	4	35	45	1.53
	5	30	37	1.25
	6	25	42	1.55
	7	20	38	1.53
	8	15	45	1.25
	9	10	37	1.55
	10	05	42	1.53

Table 2: Data from T-l curve

Time period, T ₁ (sec)	$l_1 + l_2 = (ac + bd)/2$ (cm)	g_1 (cm/sec)	Time period, T ₂ (sec)	$t_1' + t_2' = (d'e' + b'd)/2$ (cm)	g_2 (cm/sec)	Mean g (cm/sec)
1.51	24	980	1.42	22	970	975

Results:

Acceleration due to gravity, $g = 975 \text{ cm/sec}$

Percentage of error = 4 %

Precautions

1. Amplitude of oscillation was kept with 4°, otherwise the oscillation would not have been simple harmonic, which is required for the validity of Eq. 8.
2. The pendulum was carefully made to oscillate on the knife-edge for every change of hole to minimize friction and care was also taken to avoid any wobbling of the pendulum during the oscillations.

Discussion:

1. Amplitude of oscillation was kept with 4°, otherwise the oscillation would not have been simple harmonic, which is required for the validity of Eq. 8.
2. The pendulum was carefully made to oscillate on the knife-edge for every change of hole to minimize friction and care was also taken to avoid any wobbling of the pendulum during the oscillations.

8. Results:

9. Discussion



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

MARKING

Attendance	10	Marks
-------------------	-----------	--------------

Lab Report Preparation	50	Marks
-------------------------------	-----------	--------------

(Four experiments, each expt. have 12.5 Marks)

Lab Quiz/Viva Voce	40	Marks
---------------------------	-----------	--------------

TOTAL	100	Marks
--------------	------------	--------------



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

Attention

Mandatory Tools



- Pencil
- Eraser
- Scale
- Graph Paper
- Calculator



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর



**THANK
YOU !**



Department of Physics

ঢাকা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়, গাজীপুর

Any Question?